1. Budget-Aware Claim Verification for Reliable Retrieval-Augmented Question Answering

Research Area

Generative AI, retrieval-augmented generation, and factual reliability.

Existing Base Paper and Link

Base paper: *MiloNet: A Framework for Traceable and Verified RAG*

Paper link:
https://openreview.net/pdf?id=wBuEiCVBEn

Base-paper overview:
MiloNet combines evidence retrieval, controlled answer construction, and verification to address unsupported claims and missing information. The authors identify increased inference cost and limited benchmark coverage as limitations, making verification efficiency a relevant extension. (openreview.net)

Draft Abstract — Proposed Study

This proposed study will investigate budget-aware claim verification for retrieval-augmented question answering. Instead of applying the same verification process to every generated statement, the system will estimate the verification priority of individual claims using evidence support, retrieval agreement, and estimated checking cost. A scheduling policy will allocate a fixed verification budget to claims with the greatest expected benefit from additional checking. The method will be compared with unverified generation, random selective verification, and exhaustive verification under matched resource constraints. Evaluation will measure factual support, answer completeness, citation correctness, latency, and token consumption. The objective is to identify when selective verification can preserve answer reliability while reducing computational overhead.

Research Description and Proposed Advancement

The proposed contribution is a **claim-selection and verification-budget policy**, rather than a new language model.

Suggested implementation:

1. Retrieve documents and generate an initial answer.
2. Split the answer into separately checkable factual claims.
3. Estimate each claim's evidence support and verification priority.

4. Verify selected claims within a fixed budget.
5. Revise, qualify, or remove claims that remain unsupported.

Include the cost of the selection module itself when comparing efficiency.

Potential Real-World Use Case

University information assistant: A chatbot could answer questions about course requirements, deadlines, and academic procedures from official documents. The proposed policy would prioritize checking consequential details—such as eligibility conditions—rather than spending equal resources on every sentence.

Problem addressed: Balancing reliable answers with response time and computational cost.

Suggested Evaluation

Unsupported-claim rate, answer completeness, citation correctness, latency, token usage, and performance across different verification budgets.

---

2. Drift-Aware Post-Hoc Calibration of Tabular Foundation Models for Reliable Regression

Research Area

Machine learning, uncertainty estimation, and distribution shift.

Existing Base Paper and Link

Base paper: *Benchmarking Tabular Foundation Models for Conditional Density Estimation in Regression*

Paper link:
https://arxiv.org/abs/2603.26611

Base-paper overview:
The paper evaluates tabular foundation models as conditional density estimators. It identifies settings where calibration falls behind task-specific baselines and suggests post-hoc recalibration as a useful complement. (arxiv.org)

Draft Abstract — Proposed Study

This proposed study will develop a drift-aware recalibration method for predictive distributions produced by tabular foundation models. The

method will use a held-out calibration set and a limited stream of newly labeled observations to adjust predictions when deployment data differ from the original training distribution. A weighting mechanism will balance historical calibration information against recent, locally relevant observations. Experiments will compare the proposed method with uncalibrated predictions and fixed recalibration strategies under controlled distribution shifts. Evaluation will examine prediction-interval coverage, interval width, distributional prediction quality, and adaptation cost. The study will explicitly investigate the conditions under which recalibration succeeds or fails, without assuming reliability under arbitrary shifts.

Research Description and Proposed Advancement

The proposed advancement is a **shift-sensitive calibration rule** that operates after prediction.

Suggested implementation:

1. Obtain predictive distributions from a pretrained tabular model.
2. Measure changes in observable input distributions.
3. Update calibration using an explicitly defined availability of recent labels.
4. Adjust the balance between global and local calibration.
5. Test gradual shifts, abrupt shifts, and limited-label settings.

Separate calibration data from final evaluation data to prevent leakage.

Potential Real-World Use Case

Retail demand planning: A forecasting system could produce demand estimates with uncertainty intervals as customer behavior changes. The proposed method would aim to make those intervals more informative for inventory decisions.

Problem addressed: Forecast uncertainty becoming unreliable when deployment conditions change.

Suggested Evaluation

Prediction-interval coverage, average interval width, continuous ranked probability score, prediction error, adaptation time, and sensitivity to calibration-set size.

---

3. Noise-Adaptive Confidence Calibration for Deep Image Classification under Class Imbalance

Research Area

Deep learning, image classification, and confidence calibration.

Existing Base Paper and Link

Base paper: *Socrates Loss: Unifying Confidence Calibration and Classification by Leveraging the Unknown*

Paper link:
https://arxiv.org/abs/2604.12245

Base-paper overview:
Socrates Loss incorporates an auxiliary unknown class into a unified training objective for classification and confidence calibration. The proposed extension would investigate its behavior when class imbalance and label noise occur together. (arxiv.org)

Draft Abstract — Proposed Study

This proposed study will investigate confidence calibration in image classification under simultaneous label noise and class imbalance. Building on an uncertainty-aware classification objective, the method will adapt its uncertainty penalty using estimates of label reliability and class frequency. A central design goal will be to distinguish potentially mislabeled examples from difficult but valid minority-class examples. Experiments will introduce controlled noise and imbalance separately and jointly, with evaluation on untouched test sets. Comparisons will include standard classification objectives and the original uncertainty-aware method. Accuracy, minority-class performance, calibration error, and training stability will be measured. The study will determine whether the proposed adaptation improves reliability without disproportionately suppressing rare classes.

Research Description and Proposed Advancement

The proposed contribution is a **noise- and class-sensitive modification to the training objective**.

Suggested implementation:

1. Reproduce the base loss using a small image classifier.
2. Estimate label reliability from training dynamics or prediction consistency.

3. Incorporate class-frequency information.
4. Modify the uncertainty penalty.
5. Conduct separate ablations for noise adaptation and class balancing.

A particularly important analysis is whether the method incorrectly treats rare classes as unreliable.

Potential Real-World Use Case

Manufacturing defect inspection: An image classifier could identify uncommon defects while presenting confidence scores to a human inspector. The proposed method would aim to reduce confidently incorrect decisions when rare defects and inconsistent annotations are present.

Problem addressed: Unreliable confidence and poor recognition of underrepresented defect categories.

Suggested Evaluation

Macro-F1, minority-class recall, accuracy, expected calibration error, negative log-likelihood, and variation across repeated training runs.

---

4. Temporal and Contradiction-Aware Memory Retrieval for LLM Agents under Fixed Token Budgets

Research Area

Agentic AI, information retrieval, and long-term memory.

Existing Base Paper and Link

Base paper: *Just-In-Time Agent Memory with Runtime Agentic Research*

Paper link:
https://arxiv.org/abs/2609.34385

Base-paper overview:
Just-In-Time Agent Memory constructs query-specific context at runtime by retrieving and integrating evidence from stored histories. It provides a starting point for studying how agents should retrieve changing or contradictory information. (arxiv.org)

Draft Abstract — Proposed Study

This proposed study will develop a memory-retrieval policy that accounts for temporal validity and contradictory evidence under a fixed token

budget. Stored observations will include timestamps, source information, and links to explicit corrections where available. At query time, the system will rank evidence by relevance, authority, temporal applicability, and conflict status rather than relying on recency alone. Unresolved contradictions will trigger further retrieval or a qualified response. Evaluation will use interaction histories containing changed preferences, corrected facts, and conflicting records. Comparisons with relevance-only and recency-based retrieval will measure answer correctness, stale-information errors, contradiction handling, and resource consumption. The objective is more dependable memory use without unrestricted context expansion.

Research Description and Proposed Advancement

The proposed contribution is a **memory-selection policy that distinguishes outdated facts from unresolved disagreements**.

Suggested implementation:

1. Store individual memory items with timestamps and provenance.
2. Detect explicit updates and potentially conflicting statements.
3. Retrieve evidence according to the question's relevant time.
4. Allocate a limited context budget across supporting and conflicting evidence.
5. Stop retrieval when evidence is sufficient or the budget is exhausted.

Avoid assuming that the newest statement is always correct.

Potential Real-World Use Case

Long-running project assistant: A team assistant could track revised deadlines, changed responsibilities, and updated requirements across meeting notes. It would aim to answer from the applicable version rather than an obsolete decision.

Problem addressed: Agents relying on stale or contradictory information.

Suggested Evaluation

Answer accuracy, stale-fact error rate, contradiction-resolution accuracy, appropriate uncertainty, retrieved tokens, and latency.

---

5. Recovery-Aware Plan Verification for Defending Tool-Using LLM Agents against Indirect Prompt Injection

Research Area

Agent security, tool use, and safe recovery.

Existing Base Paper and Link

Base paper: *PlanGuard: Defending Agents against Indirect Prompt Injection via Planning-based Consistency Verification*

Paper link:
https://arxiv.org/abs/2604.10134

Base-paper overview:
PlanGuard uses an isolated planner to derive authorized actions from user instructions, then checks tool actions against constraints and user intent. This provides a foundation for investigating secure recovery when legitimate execution requires a plan change. (arxiv.org)

Draft Abstract — Proposed Study

This proposed study will investigate recovery-aware plan verification for tool-using language-model agents. The method will distinguish authorized operational recovery from plan changes suggested by untrusted external content. Following a tool failure, a constrained recovery module will propose alternatives that remain within the original user-authorized objectives, permissions, and destinations. Any expansion of authority will require explicit user approval. Experiments in a sandbox will combine ordinary tasks, tool failures, and benchmark prompt-injection cases. Comparisons will measure legitimate-task completion, false blocking, recovery success, attack success, and verification overhead. The objective is to improve operational flexibility without allowing recovery mechanisms to become an alternative route for unauthorized actions.

Research Description and Proposed Advancement

The proposed advancement is **controlled replanning that preserves the original security boundaries**.

Suggested implementation:

1. Derive task permissions and constraints from trusted instructions.
2. Verify each proposed tool action.
3. Identify operational failures such as unavailable tools.

4. Generate alternatives restricted to the original permissions.
5. Request user approval when no authorized alternative exists.

Conduct all experiments with mock services, synthetic accounts, and non-sensitive data.

Potential Real-World Use Case

Internal document-management assistant: An agent could locate and summarize approved documents. If a search service fails, it could switch to an authorized alternative without following instructions embedded in retrieved documents to disclose information elsewhere.

Problem addressed: Recovering from tool failures without weakening security controls.

Suggested Evaluation

Attack success rate, legitimate-task completion, false blocking, recovery success, unauthorized-action rate, and verification latency.

---

6. Missingness-Shift-Aware Attention for Imputation-Free Tabular Classification

Research Area

Tabular deep learning, missing data, and robustness.

Existing Base Paper and Link

Base paper: *Imputation-free Learning of Tabular Data with Missing Values using Incremental Feature Partitions in Transformer*

Open-access paper link:
https://arxiv.org/abs/2504.14610

Publisher link:
https://www.sciencedirect.com/science/article/abs/pii/S1568494626010847

Base-paper overview:
The IFIAL method uses attention masks and incremental feature partitions to learn from incomplete tabular data without explicitly imputing missing values. The proposed extension focuses on changes in missingness patterns between training and deployment. (arxiv.org)

Draft Abstract — Proposed Study

This proposed study will investigate imputation-free tabular classification when missing-data patterns change between training and deployment. An attention-based model will incorporate observable missingness indicators and feature-availability statistics to adjust how information is combined. Training will include controlled variation in missingness patterns, while evaluation will include unseen patterns and different missing-value rates. The proposed method will be compared with the original imputation-free model and representative imputation-based approaches. Evaluation will measure predictive performance, calibration, computational cost, and degradation under missingness shift. The study will distinguish robustness to missing-value quantity from robustness to changes in which features are missing and how those omissions relate to observed data.

Research Description and Proposed Advancement

The proposed contribution is an **attention mechanism or training strategy that explicitly targets train–test missingness mismatch**.

Suggested implementation:

1. Reproduce the original imputation-free model.
2. Represent which features are available for each observation.
3. Introduce varied feature-dropout patterns during training.
4. Adapt attention using observed availability information.
5. Evaluate on held-out missingness patterns.

The method should not assume that the true missingness mechanism is known at deployment.

Potential Real-World Use Case

Industrial condition monitoring: A classifier could assess equipment status when some sensors become unavailable or communication failures affect particular measurements.

Problem addressed: Maintaining useful predictions when the pattern of available measurements differs from training conditions.

Suggested Evaluation

Macro-F1, AUROC where appropriate, calibration error, performance degradation across missingness patterns, runtime, and memory use.

---

7. Stability-Aware Feature Screening with Pretrained Tabular Foundation Models

Research Area

Feature selection, tabular learning, and reproducibility.

Existing Base Paper and Link

Base paper: *PACE: Plug-and-Play Contextual Embedding for Feature Screening with Pretrained Tabular Foundation Models*

Paper link:
https://arxiv.org/abs/2609.23574

Base-paper overview:
PACE uses a frozen tabular foundation-model encoder to create contextual feature representations before feature scoring. The proposed extension would examine whether screening can become more stable across small changes in the training sample. (arxiv.org)

Draft Abstract — Proposed Study

This proposed study will develop a stability-aware feature-screening method using representations from pretrained tabular foundation models. The screening criterion will combine predictive relevance, redundancy control, and selection consistency across resampled training sets. Rather than retaining features solely on a single ranking, the method will aggregate evidence about which features remain useful across repeated samples. Evaluation will include datasets with correlated predictors, irrelevant variables, and limited training observations. Comparisons with the original embedding-based screening method and conventional selectors will measure downstream prediction quality, selection stability, feature count, and computational cost. Group-level analyses will distinguish genuinely unstable selection from interchangeable choices among highly correlated features.

Research Description and Proposed Advancement

The proposed contribution is a **feature-ranking rule that balances usefulness and reproducibility**.

Suggested implementation:

1. Compute contextual representations using a frozen encoder.
2. Score features across bootstrap or subsampled training sets.

3. Estimate selection consistency.
4. Penalize unnecessary redundancy.
5. Evaluate selected subsets using multiple downstream learners.

Report both individual-feature stability and stability of correlated feature groups.

Potential Real-World Use Case

Reducing industrial measurement requirements: A monitoring team could use the selected feature subset to investigate which measurements are consistently informative before considering sensor reduction.

Problem addressed: Feature-selection results changing substantially between similar training samples.

Important boundary: Predictive selection alone would not establish that removing a sensor is operationally safe.

Suggested Evaluation

Selection overlap, group-level stability, downstream prediction performance, selected-feature count, runtime, and sensitivity to sample size.

---

8. Corruption-Adaptive Feature Reweighting for Training-Free Open-Vocabulary Semantic Segmentation

Research Area

Computer vision, vision-language models, and robustness.

Existing Base Paper and Link

Base paper: *Improving Visual Discriminability of CLIP for Training-Free Open-Vocabulary Semantic Segmentation*

Paper link:
https://arxiv.org/abs/2510.23894

Base-paper overview:
LHT-CLIP modifies layer-, head-, and token-level processing to improve segmentation without additional training. This provides a foundation for studying whether inference-time reweighting can adapt to degraded image quality. (arxiv.org)

Draft Abstract — Proposed Study

This proposed study will investigate corruption-adaptive feature reweighting for training-free open-vocabulary semantic segmentation. A deterministic inference-time rule will estimate image-quality and feature-consistency signals, then adjust the contributions of selected layers or attention heads. The pretrained model weights will remain unchanged. Experiments will evaluate clean images and controlled corruptions, including blur, noise, and reduced illumination, across multiple severity levels. Comparisons will include the original segmentation framework and fixed reweighting strategies. Evaluation will measure segmentation accuracy, boundary quality, computational overhead, and degradation relative to clean performance. The objective is to determine whether lightweight adaptation can improve robustness without additional model training or access to test labels.

Research Description and Proposed Advancement

The proposed advancement is **input-dependent reweighting rather than one fixed configuration for every image**.

Suggested implementation:

1. Reproduce the original training-free segmentation method.
2. Compute image-quality or internal feature-consistency signals.
3. Define a lightweight rule for adjusting feature contributions.
4. Fix any chosen settings using separate development data.
5. Evaluate on unseen images and corruption severities.

Keep model training, development-set tuning, and final testing clearly distinguished.

Potential Real-World Use Case

Warehouse scene analysis: An inspection system could segment objects described by text when images are affected by dim lighting or motion blur. The proposed method would aim to preserve segmentation quality without retraining for each condition.

Problem addressed: Performance degradation when image quality changes.

Suggested Evaluation

Mean intersection-over-union, boundary quality, corruption severity curves, clean-image performance, latency, and memory consumption.

---

9. Evidence-Sufficiency Calibration for Abstention in Code-Mixed Question Answering

Research Area

Natural language processing, multilingual AI, and reliable answering.

Existing Base Paper and Link

Base paper: *Knowing When Not to Answer: Mitigating Social Bias in LLMs via Epistemic Abstention*

Paper link:
https://openreview.net/pdf/2a92ee648f3abd3305d07802e3c2221e83869020.pdf

Base-paper overview:
The paper introduces EPIK, an evidence-sufficiency framework intended to encourage abstention when context does not justify an answer. Its focus on unsupported social inferences motivates studying evidence-sufficiency decisions in code-mixed questions. (openreview.net)

Draft Abstract — Proposed Study

This proposed study will investigate evidence-sufficiency calibration for question answering in code-mixed language settings. The system will assess whether a supplied context supports an answer and choose among answering, requesting clarification, or abstaining. A bilingual evaluation set will contain matched answerable and unanswerable questions with varying degrees of code-switching. Calibration will use a separate development split, and the study will compare direct answering, prompted abstention, and an evidence-sufficiency baseline. Evaluation will measure correctness among answered questions, answer coverage, unsupported-answer frequency, and unnecessary abstention. The objective is to improve evidence-sensitive decision-making without treating linguistic variation itself as a reason to refuse an answer.

Research Description and Proposed Advancement

The proposed contribution is a **calibrated evidence-sufficiency decision for code-mixed inputs**, not merely a translated dataset.

Suggested implementation:

1. Select a language pair that can be reliably annotated.
2. Create matched monolingual and code-mixed questions.
3. Label evidence sufficiency independently of language style.
4. Calibrate answer-versus-abstain decisions.
5. Audit errors with bilingual reviewers.

Include tests showing whether the system unnecessarily refuses questions from particular language groups.

Potential Real-World Use Case

Multilingual university help desk: Students could ask questions using mixed languages. When the available handbook does not contain the answer, the assistant could request missing details or refer the question to staff instead of inventing a response.

Problem addressed: Unsupported answers and excessive refusal in multilingual interactions.

Suggested Evaluation

Answer coverage, accuracy among answered questions, risk–coverage curves, false-abstention rate, unsupported-answer rate, and performance by language condition.

---

10. Failure-Type-Aware Recovery Policies for LLM Agents under Tool-Call Constraints

Research Area

Agentic AI, planning, and resource-aware recovery.

Existing Base Paper and Link

Base paper: *Agent Planning Benchmark: A Diagnostic Framework for Planning Capabilities in LLM Agents*

Paper link:
https://arxiv.org/abs/2606.04874

Base-paper overview:
Agent Planning Benchmark separates planning diagnosis from execution outcomes and includes broken tools, irrelevant tools, and unsolvable

tasks. Its updated version also evaluates diagnostic refinement, which should be included when assessing a proposed recovery policy. (arxiv.org)

Draft Abstract — Proposed Study

This proposed study will develop a failure-type-aware recovery controller for language-model agents operating under limited tool-call budgets. After an unsuccessful action, the controller will classify the observed failure and select among retrying, correcting parameters, substituting an authorized tool, repairing the plan, or stopping. Decisions will depend on the failure evidence, remaining budget, and estimated likelihood of useful progress. Evaluation will combine planning diagnostics with executable sandbox tasks containing controlled failures. Comparisons will include fixed retries, full replanning, and diagnostic refinement under matched resource budgets. The study will measure completion, recovery efficiency, unnecessary calls, and appropriate stopping, with the goal of identifying when targeted recovery outperforms uniform retry strategies.

Research Description and Proposed Advancement

The proposed contribution is a **mapping from failure diagnosis and remaining budget to an appropriate recovery action**.

Suggested implementation:

1. Define a small, observable failure taxonomy.
2. Collect or generate sandbox execution traces.
3. Implement a rule-based controller as an initial baseline.
4. Develop and compare a learned or calibrated recovery policy.
5. Test unseen task templates and failure combinations.

Evaluate actual execution, not only whether a generated recovery plan appears reasonable.

Potential Real-World Use Case

IT service-desk assistant: An agent could collect diagnostic information from authorized internal tools. If one tool times out, it could retry selectively, use an approved alternative, or escalate rather than repeat ineffective calls.

Problem addressed: Wasted tool calls and ineffective recovery during automated workflows.

Suggested Evaluation

Task completion, recovery success, tool calls per completed task, unnecessary retries, stopping accuracy, latency, and total computational cost.